

More Gauss's Law — the Three Symmetries (completed)

PHYS 235 · Wed Sep 23 · Knight 24.4–24.6

Goal. State Gauss's law and understand what it says, then use it to *compute* $\vec{E}$ — not just check it — for the distributions where symmetry makes that possible: a point charge and a shell, an infinite line, an infinite sheet, and a uniformly charged sphere.

1. What Gauss's law says

a) Flux. The number of electric field lines that exit or enter a **closed surface** is the same, no matter the **size or shape** of the surface. By “number of field lines” we mean *flux*:

$$\Phi = \oint \vec{E} \cdot d\vec{A}$$

Notes: the surface's area vector is **perpendicular** to the surface and points **outward**, by convention. If lines are exiting, Φ is **positive**; if entering, Φ is **negative**. A line that just passes through counts +1 for leaving and −1 for entering, so it contributes **zero** net.

b) Relation to charge. The number of field lines — also called **net flux** — through a closed surface is equal to $q_{\text{enc}}/\epsilon_0$. Does it matter how far from the charge the surface is? **No**. The closed surface we integrate over is called a **Gaussian surface**.

Check. A point charge sits at the center of a Gaussian sphere. If you double the sphere's radius, does Φ change? Does E at the surface change? (Different questions!)

Φ doesn't change (same enclosed charge). E *does* change — it drops by a factor of 4, since $E = q/(4\pi\epsilon_0 r^2)$ even though the total flux through the bigger sphere is the same.

2. Using it: match the surface to the symmetry

a) Use Gauss's law when the charge distribution has one of the three classical symmetries: **spherical**, **cylindrical**, **planar**.

Definition of symmetry: If you closed your eyes while the charge distribution changed in a symmetric way, then opened them again, you couldn't tell anything had changed.

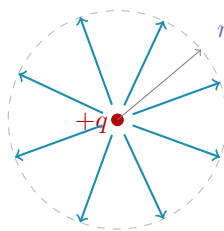
b) Choose a Gaussian surface that respects the **symmetry** of the charge distribution. If the distribution is spherically symmetric, make your surface spherically symmetric too, **centered on the charge distribution** (similarly for cylindrical and planar symmetry).

If you've done this, $\vec{E}$ will be everywhere either **parallel** or **perpendicular** to your Gaussian surface, and $\Phi = E \oint dA = EA$ — Gauss's law becomes plain algebra: $EA = q_{\text{enc}}/\epsilon_0$.

Check. Why does a cube fail as a Gaussian surface for a single point charge?

On a cube's faces $\vec{E}$ is neither constant in magnitude nor everywhere parallel/perpendicular to the face — both the distance to the charge and the angle between $\vec{E}$ and $d\vec{A}$ vary across each face, so E can't be pulled out of the integral.

3. Warm-up: a point charge, then a shell



By symmetry $\vec{E}$ is radial and constant in magnitude on the sphere, so $EA = q_{\text{enc}}/\epsilon_0$ with $A = 4\pi r^2$ and $q_{\text{enc}} = q$ gives $E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$ — Coulomb's law, recovered from Gauss's law.

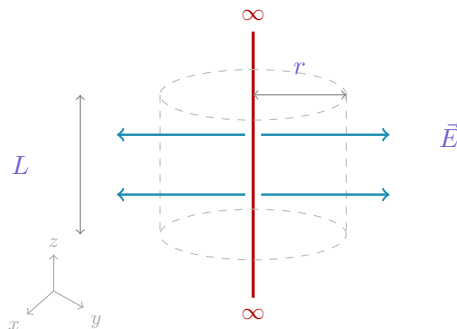
Now make it a thin **shell** of charge Q , radius R , instead of a point. **Outside** ($r > R$): the Gaussian sphere encloses Q (all of it), so $E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$ (identical to a point charge — the shell might as well not have a size). **Inside** ($r < R$): the Gaussian sphere encloses **0**, so $E = 0$.

Check. The sphere you'll work below is *solid*, not a shell — charge fills the whole volume. Will its field inside behave the same way as the shell's?

No — a solid sphere **DOES** enclose charge inside radius r (it grows with r^3), so unlike the shell, $E \neq 0$ inside. See the result below: $E \propto r$.

4. Infinite line of charge

Linear charge density λ . $\vec{E}$ points radially out from the wire and cannot depend on position along it, so the Gaussian surface is a *coaxial cylinder* of radius r and length L .



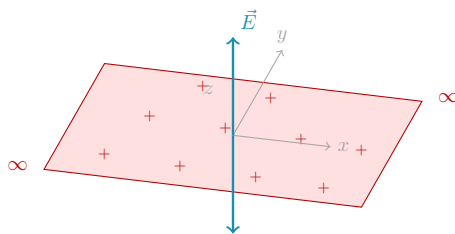
Which part of the cylinder carries flux (and which carries none)? Area A , then q_{enc} , then solve for E . Show L cancelling.

Only the curved side carries flux ($\vec{E} \parallel d\vec{A}$ there); the flat end caps carry none ($\vec{E} \perp d\vec{A}$). $A_{\text{side}} = 2\pi rL$, $q_{\text{enc}} = \lambda L$, so $E(2\pi rL) = \lambda L/\epsilon_0$ — L cancels.

E for an infinite line $E = \frac{\lambda}{2\pi\epsilon_0 r}$

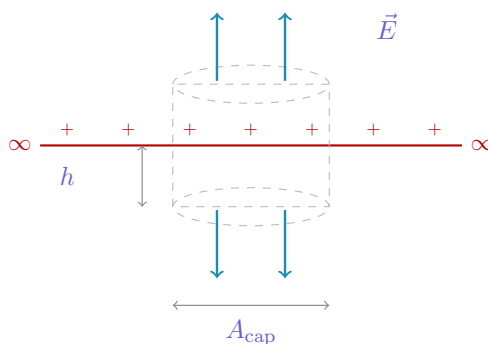
5. Infinite sheet of charge

Surface charge density η . This is a *plane*, not a line: it extends forever in *two* directions, x and y , and $\vec{E}$ points straight away from it along $\pm z$ on both sides.



An infinite sheet, seen at an angle: it extends forever in x and y , and $\vec{E}$ points straight out along z on either side.

Now look at it *edge-on* (sighting along y , so the sheet appears as a line and the pillbox is easy to draw), with end caps of area A_{cap} :



Total area carrying flux, q_{enc} in terms of η and A_{cap} , then E .

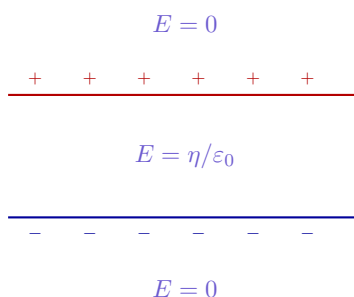
Both caps carry flux, total area $2A_{\text{cap}}$; $q_{\text{enc}} = \eta A_{\text{cap}}$; so $E(2A_{\text{cap}}) = \eta A_{\text{cap}}/\epsilon_0$ — A_{cap} cancels.

E for an infinite sheet $E = \frac{\eta}{2\epsilon_0}$

Check. Two surprises. A_{cap} cancelled — why did it *have* to? And the answer contains no r at all. What does that say about walking away from the sheet?

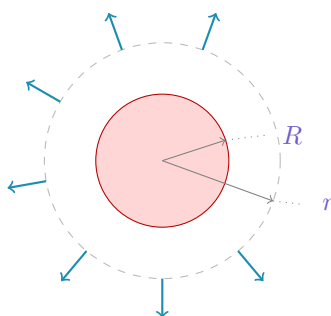
A_{cap} had to cancel because both Φ and q_{enc} scale with it equally — E can't depend on the size of a Gaussian surface you chose arbitrarily. No r : the field is the *same* at any distance — the sheet looks “infinite” from any distance you could walk to.

Then: two parallel sheets, one $+\eta$ and one $-\eta$. Superpose the two fields you just found — nothing new to derive.



6. Uniformly charged sphere

Total charge Q spread uniformly through a solid sphere of radius R . There are two regions, and the only difference between them is q_{enc} .



Outside ($r > R$): the Gaussian sphere encloses **all** Q of the charge.

q_{enc} , then E . What familiar result does this match?

$q_{\text{enc}} = Q$, so $E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$ — matches a point charge (and the shell above).

E outside the sphere $E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$

Inside ($r < R$): only the charge within radius r counts. The charge fills the **volume**, so the fraction enclosed is the ratio of **volumes**, r^3/R^3 .

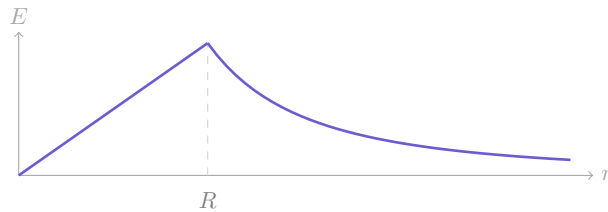
$q_{\text{enc}} = ?$ then $E = ?$

$q_{\text{enc}} = Q(r/R)^3$, so $E = \frac{1}{4\pi\epsilon_0} \frac{Qr}{R^3}$ — linear in r .

E inside the sphere $E = \frac{1}{4\pi\epsilon_0} \frac{Qr}{R^3}$

Check. Sketch E against r across both regions. At $r = R$ — jump, kink, or neither?

Neither a jump nor a gap — E is continuous at $r = R$ (both formulas agree there). But there IS a kink: the slope changes from linear ($E \propto r$) to falling ($E \propto 1/r^2$).



All three results are in the *Applying Gauss's Law* sim — open its **Info** panel to check yours.